

HASHMAP 2



Good

Morning

Content

01. check pair with given sum
02. Count pair sum
03. subarr sum = k
04. Distinct char in window of size k

01. Given N array elements, check if there exists a pair (i, j) such that $ar[i] + ar[j] = k$ & $(i \neq j)$

A[] :	8	9	1	-2	4	5	11	-6	7	5
	0	1	2	3	4	5	6	7	8	9

$k = 11 \longrightarrow ar[4] + ar[8] \quad \text{True}$

$k = 6 \longrightarrow ar[2] + ar[5] \quad \text{True}$

$k = 22 \longrightarrow \text{False}$

$A[] = \{ 3 \quad 5 \quad 1 \quad 2 \quad 1 \quad 2 \}$
 0 1 2 3 4 5

$k = 7 \rightarrow ar[1] + ar[3] = 7 \quad \text{True}$

$k = 10 \rightarrow \text{False}$

Brute force $\rightarrow$ Generate all pairs & check if pairsum = k

```
for (i=0; i<n; i++){
```

```
    x = A[i], y = k - A[i]
```

```
    for(j=i+1; j<n; j++){
```

```
        if (A[j] == y) return true;
```

```
    }
```

```
}
```

```
return false
```

TC: $O(n^2)$

SC: $O(1)$

* Approach 2 (Use hashset)

Insert all the elements inside hashset & then look for tar

$A[] = \{ \underset{0}{8} \underset{1}{9} \underset{2}{2} \underset{3}{-2} \underset{4}{4} \underset{5}{5} \underset{6}{11} \underset{7}{-6} \underset{8}{4} \}$ $K=9$

$i = \checkmark \checkmark \checkmark \checkmark$

$tar = 1 \quad 0 \quad 7 \quad 11$

$\times \quad \times \quad \times \quad \checkmark$

return true.

hs

8	9	2	-2
4	5	11	-6

* Issue

$A[] = \{ \underset{0}{8} \underset{1}{9} \underset{2}{2} \underset{3}{-2} \underset{4}{4} \underset{5}{5} \underset{6}{11} \underset{7}{-6} \underset{8}{4} \}$ $K=4$

$i = \checkmark \checkmark \checkmark$

$tar = -4 \quad -5 \quad 2 : \text{return true}$

$(K - A[i])$

$\times \quad \times$

$\times$
Not the correct

solution as $i=j$

hs

8	9	2	-2
4	5	11	-6

* HM $\rightarrow$ {Try by yourself}

* Resolve this using hashset

$\rightarrow$ put elements inside hashset from

0 to $i-1$

$A[] = \{ 8 \ 9 \ 2 \ -2 \ 4 \ 5 \ 4 \ -6 \ 11 \}$

$K = 8$

i ✓ ✓ ✓ ✓ ✓ ✓ ✓

tar = 0 -1 6 10 4 3 4

return true;

hs		
8	9	2
-2	4	5

bool pairsum (int []A, int K)

HashSet<I> set = new HashSet<>();

for (i=0 : i<n; i++) {

 x = A[i] , tar = K - x

 if (set.contains(tar)) return true;

 set.add(x);

return false

Count pair sum

Given an arr[] & an integer B. Count no. of pairs (i, j)

such that $A[i] + A[j] = B$ and $i \neq j$

Note:- The pair (i, j) is same as pair (j, i) &

should be counted once.

✓ $A = \{ \overset{0}{3}, \overset{1}{5}, \overset{2}{1}, \overset{3}{2} \}$ $B = 8$ pair (0,1) Ans = 1

$A = \{ \underset{0}{1}, \underset{1}{2}, \underset{2}{1}, \underset{3}{2} \}$ $B = 3$ pairs
 (0,1) (0,3)
 (1,2) (2,3) Ans = 4

$A[] = \{ \underset{0}{3} \underset{1}{5} \underset{2}{1} \underset{3}{2} \underset{4}{1} \underset{5}{2} \}$ $K = 3$ Ans = 4

pair = (2,3) (2,5)
 (3,4) (4,5)

$A[] = \{ 2 \ 5 \ 2 \ 5 \ 8 \ 5 \ 2 \ 8 \}$ $B = 10$

target = 8 5 8 5 2 5 8 2
 (K - A[i])

count = 0 + 0 + 0 + 1 + 2 + 2 + 1 + 3 = Ans = 9

HM = store freq of distinct
 ele from 0 to (i-1)

HM

$$\left\{ \begin{array}{l} 2 \rightarrow 3 \\ 5 \rightarrow 2 \\ 8 \rightarrow 1 \end{array} \right.$$

```
HashMap < I, I> map = new HashMap<>();
```

```
ans = 0
```

```
for (i = 0; i < n; i++) {
```

```
    x = A[i]
```

```
    tar = B - x
```

TC: $O(n)$

SC: $O(n)$

```
    if (map.containsKey(tar) == true) {
```

```
        ans += map.get(tar);
```

```
        map.put(x, map.getOrDefault(x, 0) + 1);
```

```
    }
```

```
return ans
```

03. Subarr with given sum

Given an arr[] of integers A & an integer K
check if there exist a subarr which
adds up to K

arr[] =

8	9	2	-2	4	5	11	-6
0	1	2	3	4	5	6	7

 K = 9

$$A = \{5 \ 10 \ 20 \ 100 \ 105\} \quad k = 110$$

Ans = false

$$\text{subarr sum} = \text{Pf}[r] - \text{Pf}[l-1] = k$$

$$\underbrace{\text{Pf}[a]}_x - \underbrace{\text{Pf}[b]}_y = k$$

$$x = \text{Pf}[a]$$

$$y = x - k$$

$$A[] = \{2 \ 3 \ 9 \ -4 \ 1 \ 5 \ 6 \ 2 \ 5\} \quad k = 11$$

$$\text{Pf}[] = \{2 \ 5 \ 14 \ 10 \ 11 \ 16 \ 22 \ 24 \ 29\}$$

$$\text{tar} = -9 \ -6 \ 3 \ -1 \ 0 \ 5$$

true

hs

2 , 5 , 14
10 , 11

```
int sum = 0
```

```
HashSet<I> set = new HashSet<>();
```

```
set.add(0);
```

```
for (i=0; i<n; i++) {
```

```
    sum = sum + A[i]
```

②

```
    x = sum
```

```
    y = x - k;
```

```
    if (set.contains(y) == true) return true;
```

```
    set.add(x);
```

```
}
```

```
return false;
```

8:17 AM → 8:27 AM

04. Given a string, calculate no. of distinct characters in every window of size k.

str = { a b c d a d b d d } k=4
0 1 2 3 4 5 6 7 8

substrings: $[0 \ 3] = 4$

$[1 \ 4] = 4$

$[2 \ 5] = 3$

$[3 \ 6] = 3$

$[4 \ 7] = 3$

$[5 \ 8] = 2$

* Freq array of size 26 $\rightarrow$ SC: $O(1)$

1	1	1				
0	1	2	3	4	5	6...25
a	b	c	d	e	f	g....z

$\text{freq}[\text{ch} - 'a']$

* Idea

$\rightarrow$ Create a freq array of first window & then slide the window on RHS

```
int (*) freq = new int [26];
```

```
for (i = 0; i < k; i++) {
```

```
    char ch = str[i];
```

```
    freq[ch - 'a'] ++;
```

```
}
```

```
print(countdistinct(freq));
```

```
s = 1, e = k
```

```
while (e < n) {
```

```
    char drop = str[s - 1];
```

```
    freq[drop - 'a'] --;
```

```
    char gain = str[e];
```

```
    freq[gain - 'a'] ++;
```

```
    print(countdistinct(freq));
```

```
    s ++;
```

```
    e ++;
```

```
}
```

TC: $O(n)$

SC: $O(26) = O(1)$

```
int countDistinct (int [] freq)
```

```
    count = 0
```

```
    for (i = 0; i < 26; i++) {
```

```
        if (freq[i] > 0) count ++;
```

```
    }  
    return count;
```

* Characters = A to Z & a to z
 65 to 90 97 to 122

freq[] =

123

freq[ch] ++;

freq[drop] --;

$$* \quad \boxed{j-i \leq B} \quad \& \quad A[i] = A[j] \quad j > i$$

count of pairs

$$\underline{\underline{B+1}}$$

$$A[] = \{ \overset{0}{1} \quad \overset{1}{2} \quad \overset{2}{\textcircled{1}} \quad \overset{3}{3} \quad \overset{4}{1} \quad \overset{5}{4} \} \quad B=2$$

HM

$$\left\{ \begin{array}{l} \text{Ele} \quad \text{Freq} \\ 1 \rightarrow 1 \\ \textcircled{2} \rightarrow 1 \\ 3 \rightarrow 1 \\ 4 \rightarrow 1 \end{array} \right\}$$

$$\text{ans} = 1 + 1 = 2$$

$$A[] = \{ \underset{0}{1} \quad \underset{1}{1} \quad \underset{2}{\downarrow 1} \quad \underset{3}{1} \quad \underset{4}{1} \} \quad B=2$$

HM

$$\left\{ 1 \rightarrow 1 \right\}$$

$$\text{ans} = 1 + 1 = 2$$